माधव प्रौद्योगिकी एवं विज्ञान संस्थान, ग्वालियर
MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR
Deemed to be University
(Declared under Distinct Category by Ministry of Education, Government of India)
NAAC ACCREDITED WITH A++ Grade
Gola Ka Mandir, Gwalior (M.P.)- 474005, INDIA
Ph.:+91-751-2409300, E-mail: vicechancellor@mitsgwalior.in, Website: www.mitsgwalior.in

Confidence-Aware RAG Pipeline with Hallucination Detection

A chatbot that answers only when its documents back it up.

Shivam Gupta (BTCD24O1060)

Cornerstone Project · Project Expo 2026

Artificial Intelligence & Machine Learning Systems · July–Dec 2026

The problem: chatbots make things up



Answers anyway

The documents don't cover the question, so the model fills the gap from memory.



Adds details

The right text is found, but the answer includes facts that aren't in it.



Contradicts

The answer says the opposite of what the documents say.

And it sounds just as confident either way. Users can't tell a right answer from a made-up one, and a confident wrong answer is worse than "I don't know".

My idea: check twice, then decide



Gate A · before answering

Are the retrieved passages actually about this question?

If not, stop. The language model never even runs.



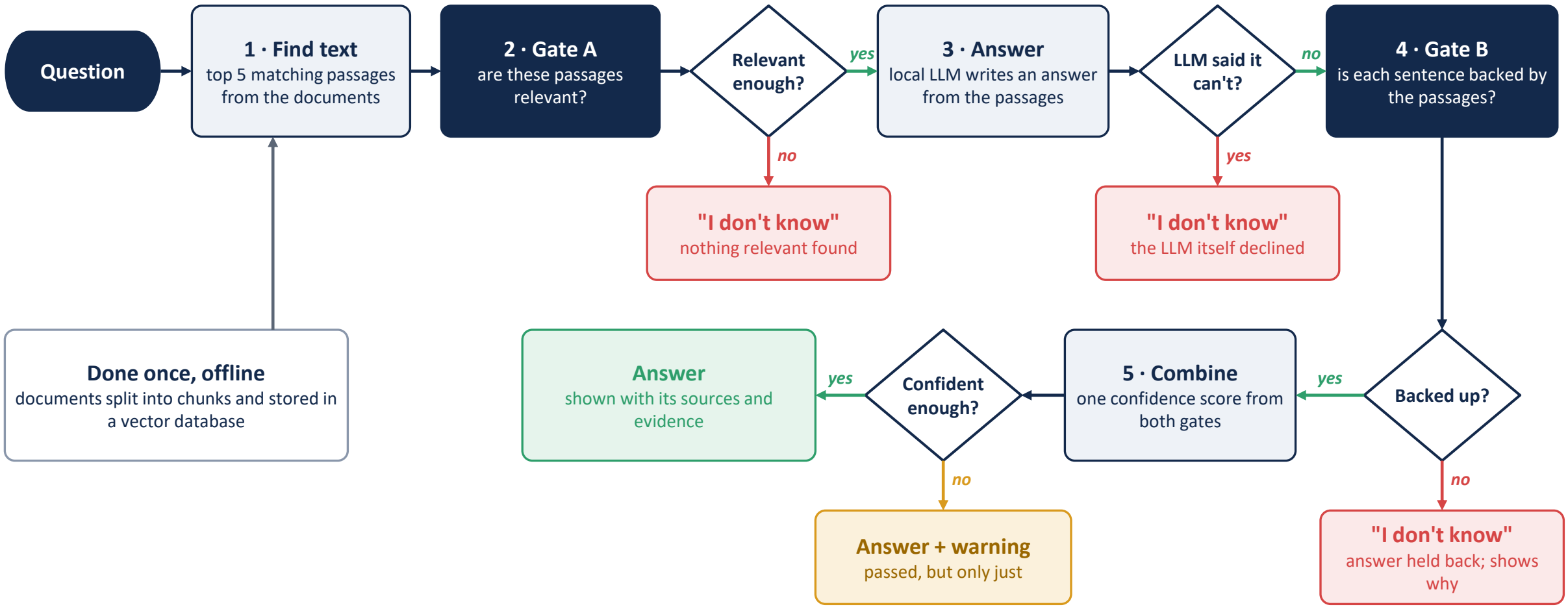
Gate B · after answering

Is every sentence of the answer backed by the passages?

If not, hold the answer back and show which sentence failed.

Pass both checks → answer, with the evidence. Fail either → "I don't know", with the reason. Runs fully offline on a laptop.

How it works: one question, step by step



Every decision is logged and shown on a monitoring dashboard.

Three choices that make it work

Two different models

Gate A ranks passages by relevance. Gate B checks if a sentence follows from a passage. Different jobs, so different models.

One weak check fails all

If relevance is 0.95 but support is 0.05:

0.50 average: would pass

0.08 my score: correctly fails

One lie isn't averaged away

If any sentence contradicts the documents, the whole answer fails, even if every other sentence is true.

How I tested it



150 test questions

From SQuAD 2.0. Half of them have no answer in the text: they are traps designed to look answerable.

Two systems, same questions, same model

A normal RAG chatbot
always answers

My system
both checks switched on

What I counted

- Made-up answers: answering a trap question (lower is better)
- Missed answers: refusing a question it could have answered (lower is better)
- Also tested on 150 harder multi-step questions (HotpotQA)

The journey: five milestones

M1

Build

The pipeline, both checks, the app

Working app, 31 tests

M2

Test

150 questions vs a normal chatbot

Made-up answers 75 → 3, but too cautious

M3

Tune

Set the check thresholds from data

Slightly less cautious

M4

Stress-test

Harder multi-step questions

Gate B struggles

M5

Audit & fix

Re-checked everything, fixed Gate B

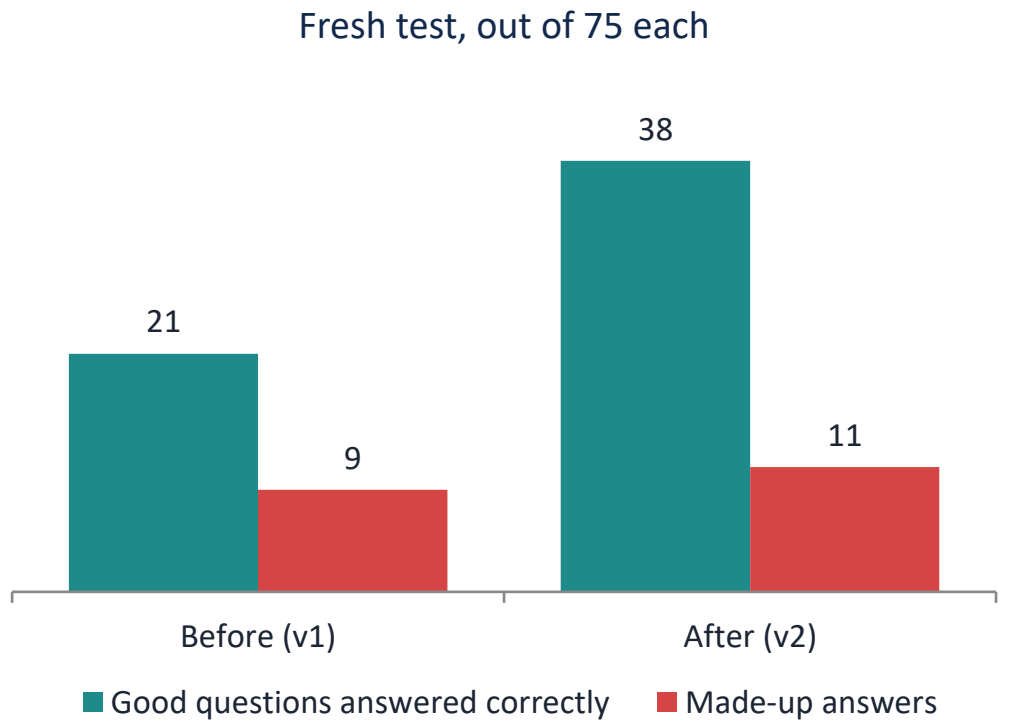
Missed answers 71% → 48%

What I found and fixed

A bug in my own scoring
Answers ending in a citation like "[1]" were marked wrong. When the system answered, it was actually right 20 of 21 times.

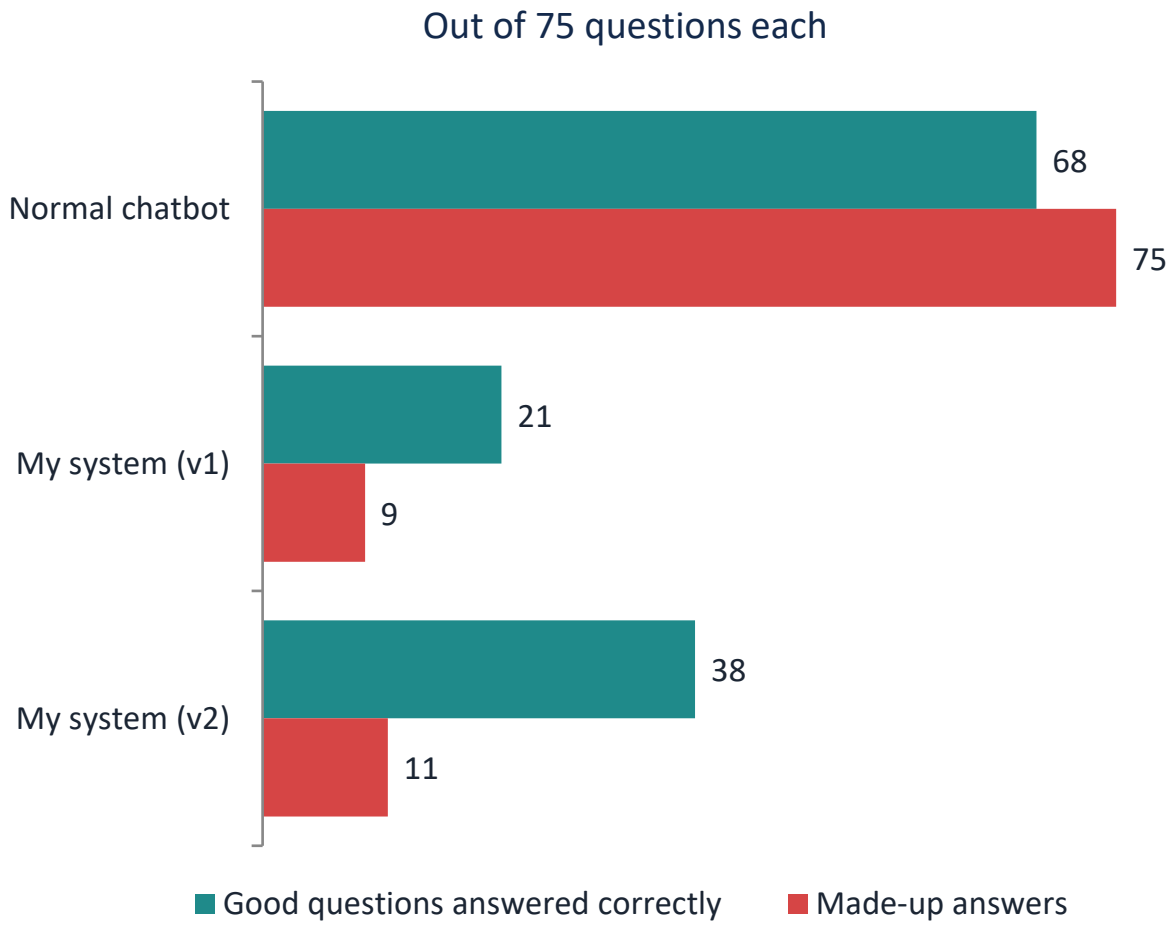
The real reason it was too cautious
Gate B rejected correct answers (35 of 36 rejections) because it read whole paragraphs at once.

The fix: Gate B v2
Check each sentence against short 3-sentence windows as well as the full passage.



17 more correct answers for 2 more mistakes, a statistically significant gain.

Results



75 → 11

made-up answers to trap questions

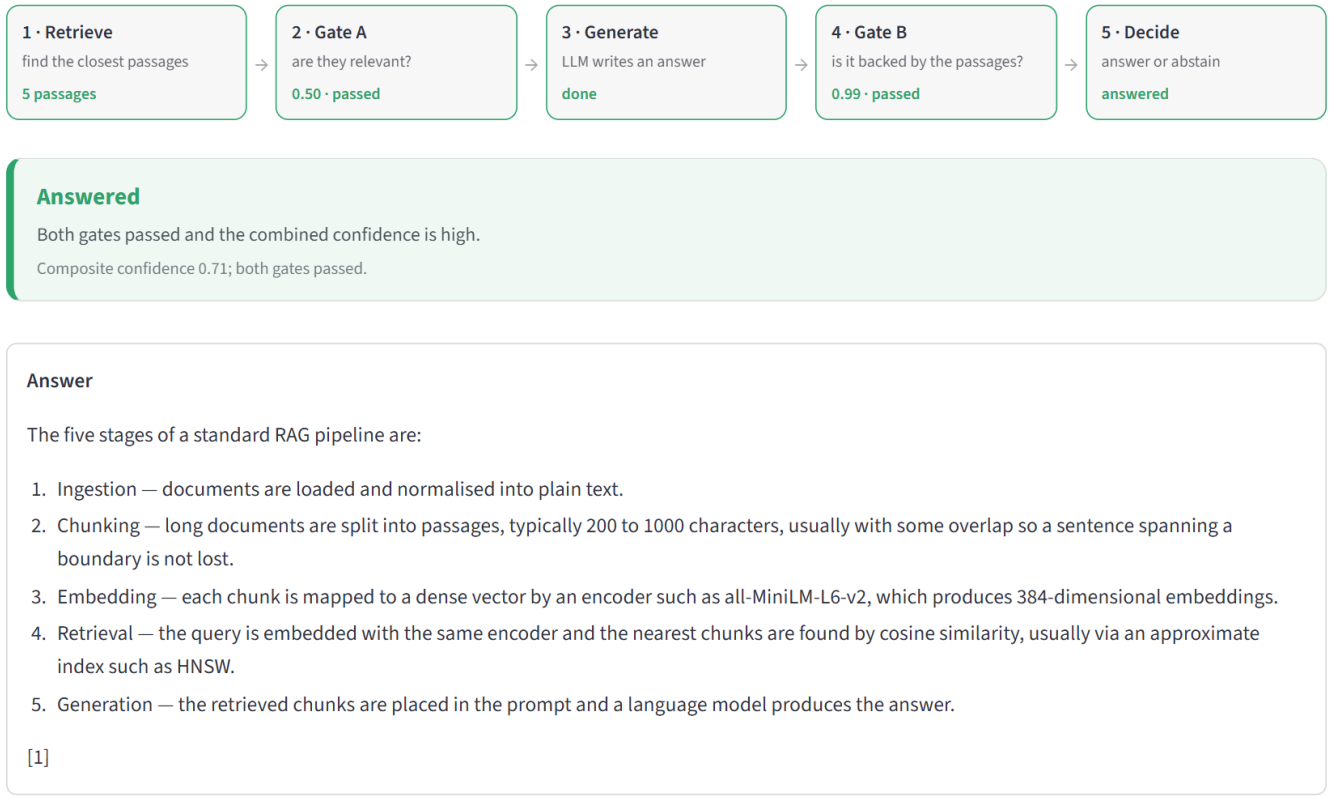
3 in 4

of its answers are correct, versus under half for a normal chatbot

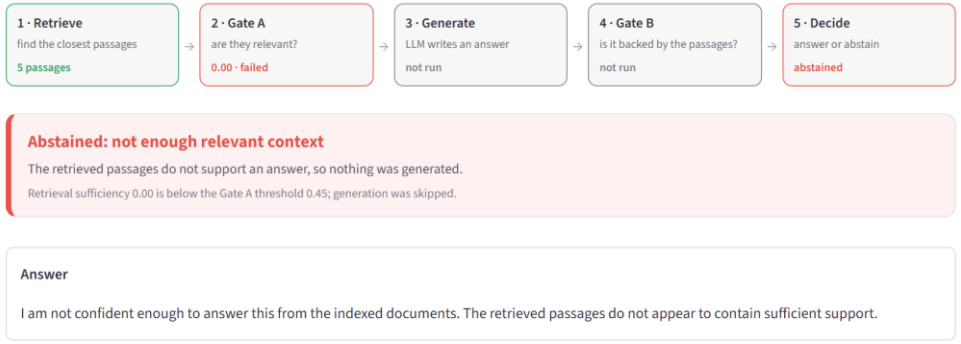
The cost

still says "I don't know" to about half of the answerable questions

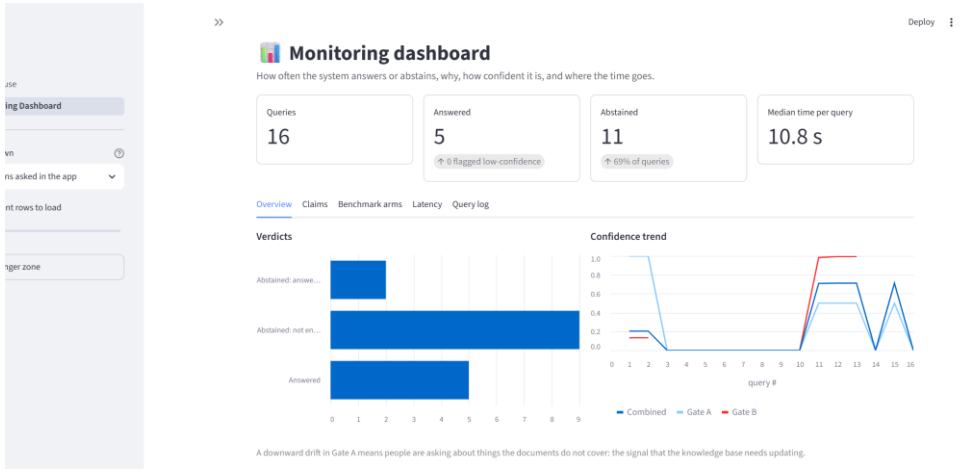
Live demo



Answered: both checks pass, evidence shown



Off-topic question: refused before the LLM runs



Dashboard: every decision logged

Limitations and what's next

Limitations

- Still too cautious: refuses about half of the answerable questions
- Gate B sometimes rejects a correct sentence
- Weak on multi-step questions, where the evidence is split across passages
- Tested on 150 questions, so the numbers have some uncertainty

What's next

- A bigger checking model: already tried, it cut wrongly rejected answers from 41% to 10% on SQuAD, but needs a bigger GPU
- Check multi-step answers against several passages together
- Test on more questions

In short

1

It knows when to say "I don't know"
Made-up answers fell from 75 to 11 out of 75.

2

It shows its reasons
Every answer or refusal comes with the evidence behind it.

3

I tested it honestly
Compared against a normal chatbot, then audited and fixed my own mistakes.

Thank you · Questions?

Shivam Gupta (BTCD24O1060) · Cornerstone Project

Backup slides

Details for questions: implementation, calibration, multi-hop results, audit.

Backup · Implementation (Milestone 1)

Layer	Files	Role
Data layer	ingest.py · embeddings.py · retriever.py	parse, chunk, embed, search
Scorers	scorers.py	reranker (Gate A), NLI (Gate B), LLM judge
Confidence layer	confidence.py	gates, contradiction cap, composite, verdicts
Generation	generator.py · llm.py	grounded prompt; Ollama, Groq, Gemini
Orchestration	pipeline.py · store.py	runs the stages, logs one row per query
Interface	app.py · pages/	app, guide, monitoring dashboard

Stack

- Python · LangChain · ChromaDB
- MiniLM embeddings + reranker
- DeBERTa-v3 NLI
- llama3.1 8B via Ollama
- Streamlit · SQLite
- 92 unit tests · ruff

Backup · Calibration and judge study (Milestone 3)

Threshold	Before	Fitted	How
Gate A	0.45	0.45	Youden's J; AUC 0.65; chosen in 185 of 200 splits
Gate B	0.60	0.25	F1, kept above the 0.15 contradiction cap
Composite	0.55	0.33	just above both gates' corner

84% → 72%

missed answers after calibration

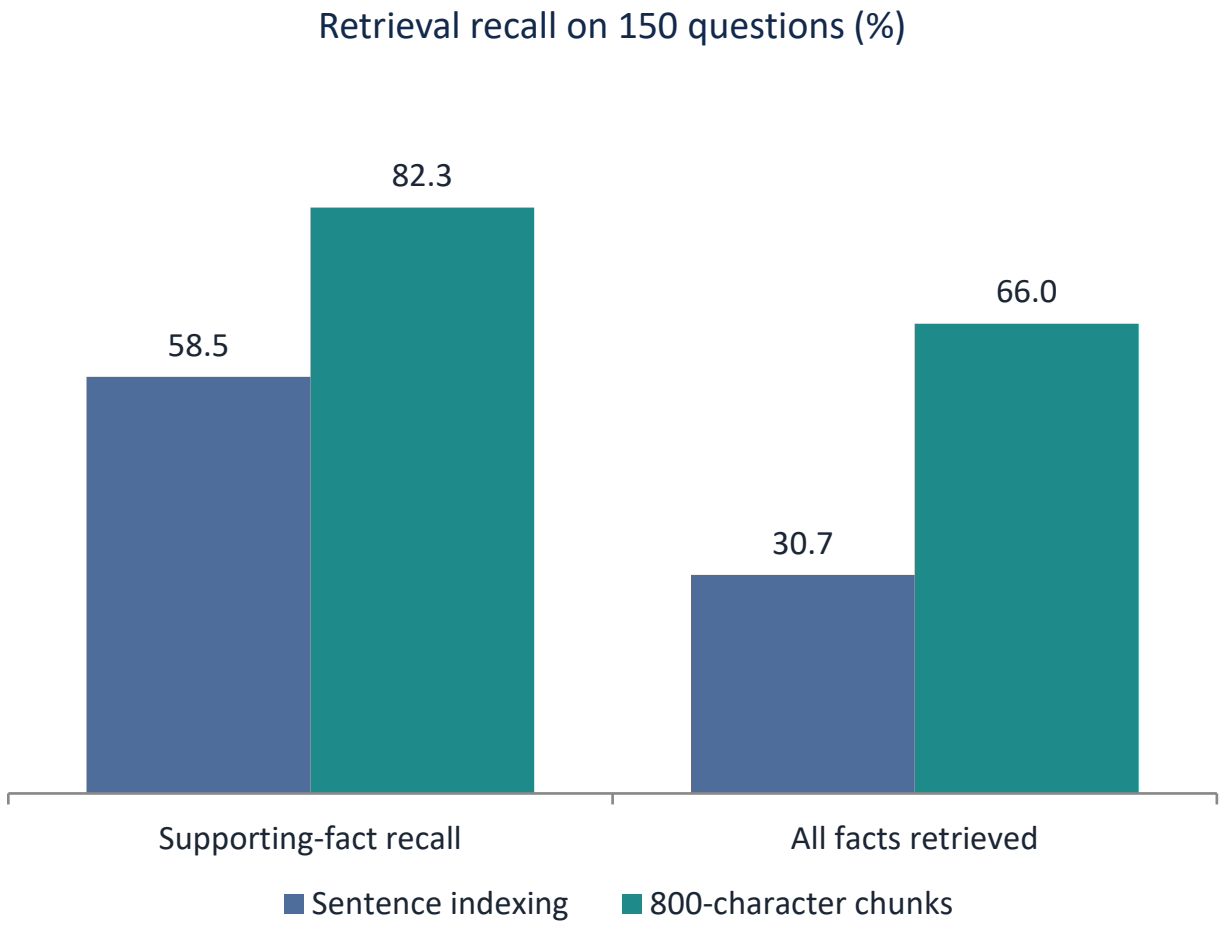
3 → 8

made-up answers of 75; p = 0.06, not significant

κ ≈ 0.04

NLI vs gemma2 LLM judge: chance-level agreement; judge ~860× slower

Backup · Multi-step questions (Milestone 4, HotpotQA)



43% → 59%

normal chatbot accuracy, sentence vs chunk indexing

41% → 24%

wrong answers among those given, with checks on (22% answered)

~2 in 3

correct multi-step answers that Gate B rejects

Backup · Audit details (Milestone 5)

Item	Finding	Result
Citation scoring	"1894 [1]" failed exact match	guarded answers right 20 of 21 times
Headline metric	aggregate EM rewards abstaining on SQuAD 2.0	report answered / abstained breakdown
Prompt vs gates	guarded prompt also allows refusing	of 67 refusals: 40 gates, 27 the model
Gate B	36 answerable answers rejected	35 were correct; fixed in v2
Gate B v2 (fresh run)	missed answers 70.7% → 48.0%	McNemar $p < 0.001$; made-up +2, $p = 0.5$
DeBERTa-large (offline)	SQuAD AUROC 0.65 → 0.70	no gain on HotpotQA; GPU memory limit